

COURSE REPORT

Assignment 1 – Task 1

Binary Classification Using K-Nearest Neighbors (KNN)

Breast Mass Diagnostic Dataset

Prepared as a formal record of the implementation, experiments, results and analysis obtained from the supplied Jupyter/Google Colab notebook.

Assignment 1 • Task 1

1. Objective and Introduction

The purpose of this task is to implement a K-Nearest Neighbors (KNN) classifier from scratch and use it for binary classification of breast mass observations. The target variable is diagnosis, with malignant observations represented by M and benign observations represented by B. The experiment is intended to demonstrate how a distance-based classifier behaves when different distance measures and neighborhood sizes are used.

Rather than relying on a pre-built KNN estimator, the supplied notebook explicitly implements the distance functions, neighbor selection, majority voting, accuracy calculation and confusion-matrix evaluation. Five distance measures are compared over five values of K. The final configuration is selected using testing accuracy, after which precision and recall are calculated for the selected model.

2. Dataset Description

The dataset loaded in the notebook contains 569 observations and 32 columns. The first columns shown in the notebook include an identifier (id), the diagnosis label, and numerical measurements such as radius, texture, perimeter, area, smoothness, compactness, concavity, concave points, symmetry and fractal dimension. These measurements are represented in mean, standard-error and worst-case groups.

The first five rows displayed by the notebook show malignant observations at the beginning of the file and contain substantially different numerical scales across the features. For example, area measurements are in the hundreds or thousands, while smoothness and fractal dimension values are much smaller. This difference in scale is important for KNN because the algorithm compares observations through distances.

The notebook creates X by dropping only the diagnosis column. Consequently, the id field is retained in X. This means that the implementation actually passes 31 columns to the classifier: the id column plus the 30 measurement attributes. This point is documented here because it is part of the actual supplied implementation.

Dataset characteristic	Value / description
Number of observations	569
Number of columns	32
Target variable	diagnosis
Target encoding	M → 1; B → 0
Features passed to KNN	31 columns in the supplied notebook, including id
Training set	455 observations (80%)
Testing set	114 observations (20%)
Random seed	42
Missing values	0 in every displayed column

3. Data Pre-processing

The preprocessing stage begins by converting the diagnosis labels into numerical form. The notebook uses NumPy to map M to 1 and all other diagnosis values to 0, so the classifier operates on binary numerical labels.

An 80:20 train-test split is then created by generating a random permutation of all 569 row indices after setting the random seed to 42. The first 455 observations are used for training and the remaining 114 observations are used for testing. The split is performed before standardization.

Standardization is performed using the mean and standard deviation calculated only from the training data. The same training statistics are then applied to both the training and test sets. In mathematical form, each feature is transformed as:

$$z = (x - \mu_{\text{train}}) / \sigma_{\text{train}}$$

This step is particularly relevant for KNN because the classifier is based directly on distance. Scaling reduces the possibility that variables with numerically larger ranges will dominate the distance calculation.

4. K-Nearest Neighbors Model

KNN is a non-parametric, instance-based classification method. The supplied implementation does not learn a set of model coefficients during training. Instead, it retains the standardized training observations and determines the class of each test observation from its nearest training examples.

For each test point, the notebook calculates the selected distance to every training point. The resulting distances are sorted in ascending order, the first K observations are retained, and their labels are collected. The predicted class is the label occurring most frequently among those K neighbors. This procedure is repeated for all 114 test observations.

The accuracy calculation is implemented as the mean of the Boolean comparison between the true and predicted labels. Therefore, an accuracy of 0.964912 corresponds to 110 correctly classified observations out of 114.

5. Distance Measures

The assignment requires five distance measures. Each one is implemented explicitly in the notebook and is selected through a common distance-calculation function.

Distance	Definition / implementation used
Euclidean	$d(x,y) = \sqrt{\sum (x_i - y_i)^2}$
Manhattan	$d(x,y) = \sum x_i - y_i $
Minkowski	$d(x,y) = (\sum x_i - y_i ^p)^{1/p}$, using $p = 3$
Cosine	$1 - [x \cdot y / (\ x\ \ y\)]$; zero-vector denominator returns distance 1
Hamming	Mean of element-wise inequality comparisons

The Hamming implementation is worth noting because the input data are continuous-valued after standardization. It measures whether corresponding numerical entries are exactly equal. For continuous floating-point features, exact equality between corresponding coordinates is uncommon, which provides a direct explanation for the much lower Hamming accuracies observed in the experiment.

6. Experimental Setup

Parameter	Setting used in the experiment
Classifier	K-Nearest Neighbors implemented from scratch
Train-test split	80% / 20%
Training observations	455
Testing observations	114
K values	3, 4, 9, 20, 47
Distance measures	Euclidean, Manhattan, Minkowski, Cosine, Hamming
Minkowski p	3
Scaling	Standardization using training-set mean and standard deviation
Selection criterion	Maximum testing accuracy
Random seed	42

The experiment therefore evaluates 25 model configurations in total (5 distance measures $\times$ 5 K values). For each configuration, all 114 test samples are classified and a single test accuracy value is recorded.

7. Results and Discussion

The notebook produced the following testing accuracies. The results are reproduced directly from the experiment and form the main basis for selecting the final KNN configuration.

Distance	K = 3	K = 4	K = 9	K = 20	K = 47
Euclidean	94.74%	92.98%	92.98%	93.86%	93.86%
Manhattan	96.49%	93.86%	93.86%	93.86%	93.86%
Minkowski	93.86%	92.98%	92.98%	93.86%	93.86%
Cosine	96.49%	95.61%	94.74%	95.61%	95.61%
Hamming	68.42%	61.40%	65.79%	64.04%	58.77%

7.1 Accuracy Trend

The following figure is the graph generated by the supplied notebook. It compares K against testing accuracy for all five distance measures. The strongest performance occurs at K = 3 for both Manhattan and Cosine distance. In contrast, Hamming remains well below the other distance measures for every tested K.

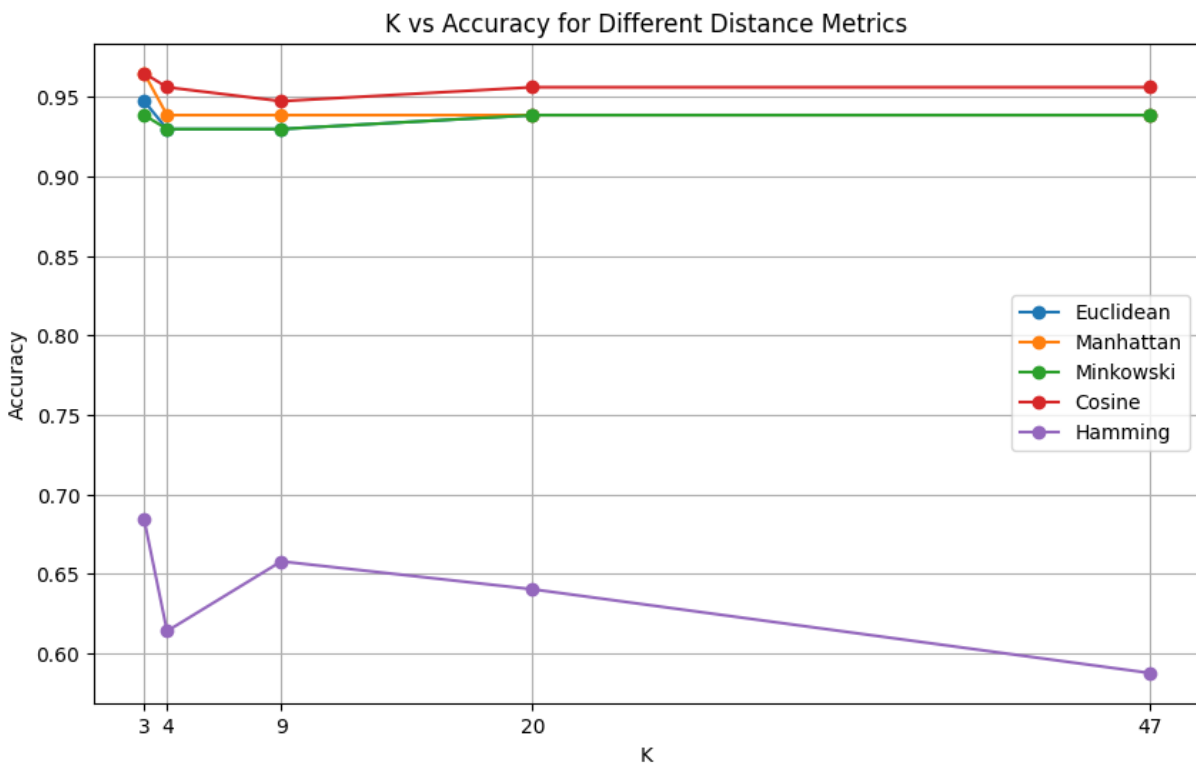


Figure 1. K versus Accuracy for Different Distance Metrics — generated directly by the supplied notebook.

7.2 Comparison by Distance Measure

Distance	Mean accuracy across K	Best accuracy	Best K	Range
Euclidean	93.68%	94.74%	3	92.98–94.74%
Manhattan	94.39%	96.49%	3	93.86–96.49%
Minkowski	93.51%	93.86%	3 / 20 / 47	92.98–93.86%
Cosine	95.61%	96.49%	3	94.74–96.49%
Hamming	63.68%	68.42%	3	58.77–68.42%

The mean values in the table are calculated from the five reported test accuracies for each metric. Cosine has the highest mean accuracy across the tested K values at approximately 95.61%, while Manhattan has the highest individual accuracy at 96.49%. This indicates that Cosine was the most consistently strong metric across the tested K values, whereas Manhattan provided the selected peak result.

7.3 Selection of the Best Model

The notebook identifies Manhattan distance with $K = 3$ as the best result because it is the first maximum returned when the results table is searched for the highest accuracy. Its testing accuracy is 0.964912, or 96.49%. Cosine distance with $K = 3$ reaches exactly the same accuracy, so the two configurations are tied on the primary selection criterion.

Final model characteristic	Observed result
Distance measure	Manhattan
Number of neighbors	$K = 3$
Correct predictions	110 / 114
Incorrect predictions	4 / 114
Test accuracy	96.49%
Precision	100.00%
Recall	91.49%

8. Detailed Evaluation of the Selected Model

8.1 Confusion Matrix

The confusion matrix generated by the notebook is $[[67, 0], [4, 43]]$, using class 0 for benign and class 1 for malignant. The matrix can be read as follows:

	Predicted Benign (0)	Predicted Malignant (1)	Actual total
Actual Benign (0)	67	0	67
Actual Malignant (1)	4	43	47
Predicted total	71	43	114

There are 67 true negatives and 43 true positives. No benign observation is incorrectly classified as malignant, giving zero false positives. Four malignant observations are classified as benign, giving four false negatives. The 110 correct predictions out of 114 test observations produce the reported 96.49% accuracy.

8.2 Precision and Recall

The notebook calculates precision as $TP / (TP + FP)$. With $TP = 43$ and $FP = 0$, the precision is $43 / 43 = 1.00$, or 100%. Recall is calculated as $TP / (TP + FN)$. With $TP = 43$ and $FN = 4$, the recall is $43 / 47 = 0.9149$, or 91.49%.

These results show that the selected model was extremely precise on the malignant predictions in this particular test split: every observation predicted as malignant was actually malignant. The lower recall indicates that a small number of malignant observations were missed. In a classification problem where missing positive cases matters, this distinction is important.

8.3 Additional Measures Derived from the Confusion Matrix

Measure	Derived value	Basis
Specificity	100.00%	$TN / (TN + FP)$
Negative predictive value	94.37%	$TN / (TN + FN)$
F1-score	95.56%	$2 \times \text{Precision} \times \text{Recall} / (\text{Precision} + \text{Recall})$

Balanced accuracy	95.74%	$(\text{Recall} + \text{Specificity}) / 2$
False-negative rate	8.51%	$\text{FN} / (\text{TP} + \text{FN})$

The additional measures above are not printed explicitly by the notebook; they are calculated here from the notebook's reported confusion matrix, precision and recall. They provide a slightly broader view of the selected model without changing the original assignment results.

9. Discussion and Inferences

The results demonstrate a clear difference between distance measures. Manhattan and Cosine distance perform best, while Euclidean and Minkowski remain competitive. The best individual accuracy is 96.49%, obtained at $K = 3$ by both Manhattan and Cosine. Increasing K does not improve the selected Manhattan model; after $K = 3$, its accuracy falls to 93.86% and remains constant for the remaining tested values.

Cosine distance is particularly consistent. Its five accuracies are 96.49%, 95.61%, 94.74%, 95.61% and 95.61%, giving a mean of approximately 95.61%. This is the highest average among the five distance measures. The result suggests that, for this standardized representation, the directional relationship between feature vectors is useful for distinguishing the classes.

Hamming distance behaves very differently. Its best result is only 68.42% and its lowest is 58.77%. The gap between the best Hamming result and the selected Manhattan result is 28.07 percentage points. Since the implementation compares exact equality between continuous feature values, Hamming is not a natural match for the numerical representation used here.

The confusion matrix also provides a useful practical interpretation. The selected model has zero false positives and four false negatives. Thus, although the overall accuracy is high, the model is not perfect at identifying every malignant observation. The distinction between precision and recall therefore should be considered when interpreting the final result.

10. Bonus Task: Decision Boundary

The assignment proposes visualizing a decision boundary for the best KNN configuration. The supplied notebook does not contain code that generates a decision-boundary plot, and no decision-boundary image is present among its outputs. Therefore, this section records the status of the bonus task rather than adding a plot that was not produced by the submitted code.

A decision-boundary visualization would normally require selecting two feature dimensions or applying a dimensionality-reduction method before plotting. Since the supplied implementation uses the full feature matrix, adding such a visualization would introduce an additional modelling choice beyond the code provided for Task 1.

11. Conclusion

This experiment successfully implemented a KNN classifier from scratch and evaluated it on the provided breast-mass diagnostic dataset. The procedure included label encoding, an 80:20 randomized train-test split, standardization based on training data, five distance measures and five values of K. In total, 25 configurations were evaluated.

The highest testing accuracy was 96.49%. The supplied notebook selected Manhattan distance with $K = 3$ as the final model, while Cosine distance with $K = 3$ produced the same accuracy. For the selected Manhattan model, 110 of 114 test observations were classified correctly. The resulting precision was 100% and recall was 91.49%, with four malignant cases classified as benign.

Overall, the experiment shows that the choice of distance metric and K has a substantial effect on KNN performance. Manhattan and Cosine distance were clearly more effective than Hamming for the standardized continuous feature representation used in this implementation. The results also show why accuracy alone should not be considered in isolation: the confusion matrix and recall reveal the four malignant observations that were missed.

12. Libraries Used

Library	Purpose in the notebook
pandas	Reading the CSV file and handling the tabular dataset.
numpy	Array operations, label conversion, random splitting, standardization and distance calculations.
matplotlib	Generating the K-versus-accuracy comparison plot.

The classifier itself is not imported from a machine-learning package. The notebook implements the required KNN operations directly using NumPy and basic Python constructs.

Appendix A. Complete Experimental Results

For completeness, the 25 accuracy values produced by the notebook are listed below. This table can be used to verify the selection of the best configuration.

Metric	K	Test accuracy
Euclidean	3	94.74%
Euclidean	4	92.98%
Euclidean	9	92.98%
Euclidean	20	93.86%
Euclidean	47	93.86%
Manhattan	3	96.49%
Manhattan	4	93.86%
Manhattan	9	93.86%
Manhattan	20	93.86%
Manhattan	47	93.86%
Minkowski	3	93.86%
Minkowski	4	92.98%
Minkowski	9	92.98%
Minkowski	20	93.86%
Minkowski	47	93.86%
Cosine	3	96.49%
Cosine	4	95.61%
Cosine	9	94.74%
Cosine	20	95.61%
Cosine	47	95.61%
Hamming	3	68.42%
Hamming	4	61.40%
Hamming	9	65.79%
Hamming	20	64.04%
Hamming	47	58.77%

Appendix B. Key Output from the Notebook

Output	Value
Training samples	455
Testing samples	114
Best distance	Manhattan
Best K	3
Accuracy	0.9649122807017544
Precision	1.0
Recall	0.9148936170212766
Confusion matrix	[[67, 0], [4, 43]]

End of Report

Appendix C. Extended Interpretation of the Experimental Results

The experimental table shows that the choice of K has a measurable effect on the classifier. For Manhattan distance, the highest result occurs at $K = 3$ with 96.49% accuracy, after which the accuracy decreases to 93.86%. This suggests that, for this particular train-test split, using a small local neighborhood preserves useful class-specific information that becomes less distinct when more neighbors are included.

The Cosine results are comparatively stable. The accuracy remains above 94% for every tested K value and reaches 96.49% at $K = 3$. This stability is useful because it indicates that the performance of the classifier is less sensitive to K under the cosine measure than under some of the other metrics.

Euclidean and Minkowski distance produce similar patterns. Their results remain within a narrow band around 93%, indicating that both metrics are capable of separating the classes reasonably well after standardization, but neither produces the maximum accuracy obtained by Manhattan or Cosine.

The Hamming results are substantially lower. Since the implementation applies Hamming distance to standardized continuous features, exact coordinate equality is rare. Consequently, the distance does not preserve the magnitude and geometric relationship between observations in the same way as the other continuous-valued metrics. This is an important methodological observation rather than simply a numerical difference.

Appendix D. Practical Interpretation and Limitations

The reported results should be interpreted as the performance of one fixed randomized 80:20 split with random seed 42. A different split could produce different accuracy, precision, recall and confusion-matrix values. Therefore, the result demonstrates the behaviour of the implemented model on the specified test split rather than establishing a universal performance level for the dataset.

Another implementation detail is that the notebook retains the id column when constructing the feature matrix. In a conventional machine-learning workflow, an identifier would normally be excluded because it does not represent a physical measurement. The report retains this detail because the purpose is to document the supplied implementation faithfully.

The experiment also evaluates the distance metrics directly rather than using a pre-built classifier. This makes the implementation useful for demonstrating the mechanics of KNN, including distance calculation, neighbor ranking and majority voting. At the same time, the absence of cross-validation means the selected K should be viewed as the best among the tested configurations for this particular test split.

Finally, the bonus decision-boundary visualization was not produced by the supplied notebook. A full decision-boundary plot would require choosing two dimensions or reducing the feature space, which would add an extra modelling decision. It is therefore more appropriate to report the bonus as not implemented than to introduce an unverified visualization.

End of Assignment 1 – Task 1 Report

Appendix E. Summary of Key Findings

1. The experiment evaluated 25 KNN configurations using five distance measures and five values of K.
2. Manhattan distance with $K = 3$ was selected as the best configuration by the supplied notebook, with a test accuracy of 96.49%.
3. Cosine distance with $K = 3$ achieved the same 96.49% accuracy, creating a tie on the primary accuracy criterion.
4. The selected model correctly classified 110 of 114 test observations.
5. The confusion matrix contains 67 true negatives, 43 true positives, zero false positives and four false negatives.
6. Precision is 100% and recall is 91.49%, showing excellent precision but four missed malignant observations.
7. Hamming distance produced the weakest results because it compares equality of continuous standardized feature values.
8. The K-versus-accuracy plot confirms that the strongest observed performance occurs at small K, particularly $K = 3$.

End of Report